Predicting Salinity Intrusion in the Vietnamese Mekong Delta — Updated Problem Description, Solution and Plan

Christopher Mackle

Introduction

The Vietnamese Mekong Delta is the country's core agricultural engine, making more than half of Vietnams rice. It alone helps deliver round about 90% of the national aquaculture production, alongside being the countries largest fruit growing region growing mainly Mangos, Bananas and Oranges. So disruptions of any sort can have a massive impact on the economy and the well being of the country. Its like the whole of Munster and the south of the republic of Ireland (south east tillage) dairy production worth about €6.3bn in 2024 being dependent on a river for there water supply from a delta. Then there comes the one big problem every year dry season salintity intrusion (basically saltwater moving inland through a range of river networks), which intern relatedly reaches damaging thresholds for irrigation and most importantly freshwater supply (useable water for agriculture). During the 2019 to 2020 dry season the UN assessments reported salinity arriving 10 to 20 days earlier than previous years, and up to 110km in land in some locations , putting roughly 658,558 people at risk from drought and fresh water shortages, producing large economic losses (in 2016 a loss of 5.5 trillion.

So therefore the goal of my project is to build a local reproducible workflow to predict short horizon(1 to 24hour) salinity and provide a light weight viewer for researchers and or operators. Recent dry seasons have brought earlier stronger intrusions, so short notice guidance matters and can make a real impact. The scope is offline only with cleaned per station datasets, benchmarked baselines against a gate index made using physics on key stations and ship a tidy viewer for a singe and two station inspection. The GNN work was paused because of the training time and lack of resources to speed it up.

Problem Description

Problem statement

My project will predict short horizon salinity at key monitoring stations in the Vietnamese Mekong Delta for local operators and or researchers, using offline cleaned per-station time-series data.

Why does it matter

The dry season salinity intrusion is a recurring problem or an operational threat because it can push saline water inland through a vast river system making it unsuitable or irrigation and domestic freshwater. In the 2019 2020 dry season, UN was reporting intrusion arriving 10 to 20 days earlier than the previous extreme year and reaching roughly 110km inland in some locations, putting an estimated 685558 people at risk from drought. The closest thing we have here for me (growing up around dairy farming) is if a large share of Irelands dairy heartland depended on a shared water intake network and, during a dry spell water just became unusable. Your decisions become more like reactions because you still have animals to water and maintain never mind your livelihood. In the Mekong the same operational pressure plays out across irrigation networks, nurseries and orchards. Which is why short horizon local forecasts can make a much needed bit of heads up to the people making a real difference.

Operational context and users

The Mekong Deltas farming systems rely on managed freshwater systems in the dry season, and the constaint is often water quality as much as quantity. Vietnamese monitoring and response uses the 4g/L salinity boundary (what I seen used in government forecasting and reporting anyway) because this level is widely treated as a practical threshold beyond which river water becomes unsuitable for the main crops (rice ect). In saline affected regions of the Mekong delta, some studies report canal water frequently going over 4g/L.

This becomes an immediate agricultural operational issue because the risk of salinity risk peaks during the winter to spring period and can shift quickly with tidal cycles and upstream stream water levels. The UN reporting during the 2019-2020 season 10 to 20 days earlier than the 2015-2016 extreme, with early impact reporting also highlighting agriculture and livelihoods being hit hard through water shortages damaged land. Meaning the vegetable cultivation was among the most severely affected.

For high level horticulture, the risk is not just a loss to this years yield it can have a trickle on effect to seedling and orchards which will take more than one season to recover. For example, reporting from Ben tre during the 2020 event described over 800 hectares (800 football fields)of fruit damages, which consisted of 5000 hectares of durian and rambutan as well as over 30million seedings damaged.

The forecast needs to enable agriculture first and foremost. A short horizon station forecast should help users make time sensitive decisions such as things like turning on pumps, when to open and close gates to protect upstream fesh water, irrigation scheduling and help prioritise under scarcity where the fresh water will go (to high value or vunerable areas of the business) .

Why forecasting is hard

Challenges in the Mekong Delta of such are, it being able to shift salinity levels within hours driven by its close proximity to tides and coastal boundary forcing it interacting with freshwater during the dry season. When the upstream discharge is low, the system becomes far more sensitive to the tide and small boundary changes. Ending up with sharp non-linear behaviour rather than smooth trends. Recent work using longer term hydro met data also finds very large inter annual variance and suggests that human interventions amplify the anomalies.

What makes it especially hard to forecast from monitoring stations is that the delta is heavily “ engineered”, and many of the controls (gates) aren’t in your dataset. There are lots of gates to get an idea of the vastness, the model widely used is the MIKE 11 delta model representing 542 dyke enclosed compartments and 2260 gate structures used for things like irrigation and flood management, including salinity. That's the scale of things that can be opened and closed.

Why is this important? Gate operations change the physics instantly, for instance one gate can produce different inland responses depending on the gate size and capacity. These gates are old; some leak and are not instrumented or consistently recorded a CIGAR field assessment of the 2015-2016 event reported leaks and other gate operational issues contributing to increased salinity levels and notes that when salinity rose rapidly, sluice gates were not immediately closed. These are some of the factors that the model cant see.

This is why some existing models also have difficulties:

1. Hydrodynamic models(MIKE11).
   They can are really good at representing tides and the flow of water for salinity levels. But DHIs own MIKE documentation explains to us that controlled structures are operated though control rules/conditions/actions based on sensors/thresholds/timeseries. So you have to tell the model when the gates are open and closed

2. Data driven short horizon models (regression/LSTM/ML)
   These are good at learning repeated tidal patterns when the system is behaving consistently. But small local gates with no logging become hidden latent variables, so the model has to infer it directly from water levels and it can fail during unusual or rapid changing in the behaviour of the gate.

Data Availability and Viability constraints

The system I will be making operates on cleaned per station datasets suitable for timeseries forecasting. Some of the practical constraints include:

- Gaps and irregularities
- Feature alignment across variables
- Station by station differences in things like behaviour.

Therefore my workflow will need consistent column checks, missing data handling and time based splitting to make sure the model is reproducible.

Scope and constraints

Its going to be offline only so there's no cloud dependencies or apis because of the Mekong being remote. The data unit is per station time series with shared features. A reproducible local pipeline with configuration and random seeds that will be logged for re runs. And a light weight local viewer to inspect data and compare behaviour across two or multiple stations.

Goals and requirements

Project goals

1. Operationally useful for short horizon forecasts.

Provide station level salinity forcasting for 24hour horizon that is suitable for the short notice operation and interpretation needed for pump and gate timing decisions.

2. Reproducible offline workflow
   Deliver a fully local pipeline that will run end to end offline, from the ingesting of data to the training evaluating and the exporting of forecast and plots

3. Practical baseline comparison
   Implement and evaluate a set of simple baselines so quality improvements can be quantified and justified with data, alongside the physics informed gate index LSTM.

4. Light weight inspection tool.
   Build and ship a small local viewer that allows the rapid inspection of data and comparison of 2 stations.

5. Clear documentation
   I want to have clear documentation to support the project formatted in a friendly way to the project is easy to use and understand.

Functional requirements

1. Viewer
   The system shall include a local viewer to inspect station CSV files.
   The viewer should also support single station charts and a multiple station overlay.
   The viewer loads local CSVs; to inspect forecasts, manually load the CSVs produced by the scripts.

2. Offline data input validation
   Load cleaned per station time series CSVs from local storage only.
   Check that required columns and timestamp parse ability, also missing columns, duplicate timestamps and obvious gaps via logs.

3. Feature Engineering
   Align the variables using common timestamps per station and build a lagged windows for a short horizon forecasting.
   Compute an optional physics informed gate index from boundary/tide and local water level signals for stations where those inputs exist.

4. Modelling, evaluation and export
   Train per station baselines and the gate index using time based spits to try and avoid leakage.

Success Criteria

1. Data readiness

- Cleaned presentation CSVs are produced with checks

- All data prep scripts run locally from a clean checkout and generate the same cleaned outputs consistently

2. Baseline benchmarks

- Persistence, linear/ridge and simple LSTM baselines are trained for the short horizons
- Per station metrics and plots are saved down with timestamps

3. Gate index LSTM performance

- LSTM gate model trains successfully on 4 stations with configs random seeds logged for reruns.
- For the 1 to 24hour horizon the gate index LSTM achieves lower MAE and or RSME than persistence and the simple LSTM on at least 3 out of 4 stations/
- Any where it doesn't improve I will have a short error analysis.

4. Outputs exported for inspection

- Metric tables and forecast plots are saved down locally for inclusion in the write up at the end.
- Per station outputs are exported as CSV files so they can be manually inspected and loaded into the viewer.

Expected Project Development Plan

Starting next week

Week 1

I will finish the gate index LSTM on 4 stations. Lock in the configurations and random sees so it can be rerun the same way everytime. Benchmark it again persistence, ridge linear and simple LSTM. And save the metrics, plots and forecast.

Week 2

I will tidy up the viewer so it looks clean, fix current bugs within the system and finish unit tests before the viewer.

Week 3 to 4

Once everything is tidy I will fix anything that comes back in feedback. I will then pin the version so the project will run the same way on every machine then rerun from a repo root. I will also write a short brief showing results and a brief slide/outline so I know exactly what I am presenting come April.

Week 5 to 6

I will build the presentation for April with the plots and screenshots. It will covering the project in its entirety. Being clear about risks and the GNN work which is deferred.

Week 7

I'll do all final dry runs on a fresh clone to make sure that everything runs clean. I will also finish the write up and package key tables figures and material as needed.

References

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